Multimedia stream binding for a pan-European services platform

P H Loosemore, E Sellin and S P Rana

The ability to carry stream information over a range of network types in a managed way will become an essential requirement for telecommunications network operators as future services evolve to include the transmission of audio/video and bulk data streams. This paper details the architecture of a working pan-European demonstrator offering multimedia services through the integration of stream control, as specified in the Object Management Group's (OMG's) control and management of audio/video streams, with a TINA-compliant service management environment. This CORBA-based demonstrator, known as the EURESCOM services platform (ESP), has been developed by six European telecommunications companies within EURESCOM project P715. Various modelling concepts, as defined by TINA, ODP and OMG, have been used and verified in a first prototype implementation, with connectivity provided by commercially available multimedia technology such as H.320/323 products. This EURESCOM services platform prototype is one of the first demonstrators world-wide to implement OMG's control and management of audio/video streams specification.

1. Introduction

There is currently a wide acceptance that distributed processing platforms will become the ubiquitous solution for the management of telecommunications applications. Concepts and architectures such as the Telecommunications Information Networking Architecture (TINA) [1] and the Open Distributed Processing Reference Model (RM-ODP) [2] have been in existence for some years, but it is only very recently that the enabling technologies, such as that specified by the Object Management Group (OMG) [3] in the form of Common Object Request Broker Architecture v2 (CORBA 2) compliant Object Request Brokers (ORBs), have become available. In the same way, multimedia and broadband applications are the accepted future direction for telecommunications services, but the means to deliver those services in a managed and quality-controlled way has only recently been considered. While delivery of these services may be required over PSTN and ISDN in the short-term, the general consensus is that global voice and data transmission will ultimately move to Internet protocol (IP) based networks. In addition, many applications will require the transmission of information in stream format, not only those offering audio and video services, but also where there is a need to carry bulk data from one location to another. The ability to carry stream information over a range of network types in a managed way will therefore become an essential requirement for telecommunications network operators.

BT, in collaboration with five European partners, has built a pan-European experimental platform or infrastructure based on middleware platform technologies. This project, running within the European Institute for Research and Strategic Studies in Telecommunications (EURESCOM) collaborative research programme [4], has been named the EURESCOM services platform, and is commonly known as the ESP. In particular OMG's CORBA was chosen for the middleware environment which implements key components of the TINA service architecture specification. Where products were not available the project has designed its own solutions and components. The project has a primary focus on interoperability between the heterogeneous platforms situated at six sites across Europe, and one of the main objectives is for a retailer at each site to be able to offer services to users at any other site. Initially this was effected using simple services such as a 'chat' application or 'shared whiteboard', which required only method invocations between platforms. These invocations were passed through the kernel transport network (KTN), which may be described as the 'software bus' that provides communication between the platforms using TCP/IP over Internet or ISDN2.

The project has recently implemented solutions for more advanced services based on audio/video (AV) streams working over ISDN2, which is currently the best available and the most cost-effective connectivity method between the WAN sites. A major task for this project has been to build the service management platform [5]; as this has been designed to be TINA-compliant, there was some logic for adopting parts of the TINA stream-binding model.

However, the OMG also provides a stream-binding model, and this standards-based design offers a more pragmatic service-layer view of stream control. Other workers have implemented the OMG model for an MPEG player application [6], but this was for a specific ORB, and did not consider interoperability between platforms.

From the middleware viewpoint, the essential requirement for delivery of services based on audio/video streams is the binding of stream interfaces and control of the stream flows. This is very much a service-layer view of connectivity which is not concerned with network aspects such as transport layers, signalling, network management or protocols. Both the TINA and OMG stream-binding models were considered for use with the ESP, and these are discussed later in this paper. The integration of stream binding, using the chosen model, with design and implementation details is also described, along with requirements for multiparty stream binding. Finally, some experiences from the interoperability tests are presented.

2. The EURESCOM services platform

The EURESCOM services platform is a prototype system which aims to demonstrate how a service management platform might be used to manage and control end-user service delivery to customers. The platform is based on object-oriented design and distributed processing, which are seen as the future technologies which will satisfy the system requirements for delivering new sophisticated services increasingly sourced from service providers on a global basis. Such systems will demand flexible access, management and charging mechanisms, as well as rapid service development and deployment. The ESP is seen as a basis for evaluation of the technology and will provide a platform on which future projects can build. It is described in detail elsewhere [7].

Of course there are many aspects of service management, e.g. fault, configuration, accounting, performance and security (FCAPS), and these will be investigated in future implementations. The initial project has been involved with building and deploying the platform, and implementing the standard interfaces to allow a user to access a retailer and request delivery of services. For an accurate evaluation of the platform these services must be typical of the type of multimedia services to be offered in the future.

A typical service which is available to a limited extent now, but which will become much more popular in the future, is videoconferencing. This is a ‘streams-based’ service as it requires a continuous stream of information to flow from end-point to end-point. The focus of this paper is to detail the integration of this streams-based service with the ESP. The ESP streams work has enabled BT and the other partners to gain valuable experience with the specification and implementation of distributed audiovisual systems. Currently available commercial products are used where possible, or beta versions when necessary, and modelling concepts as defined by ODP, OMG and TINA, such as binding object and stream interface, are used as a basis for the implementation. A key feature of the work is to investigate interoperability between the different platforms, allowing an insight into potential future problems when PNOs will be required to present open interfaces to service providers. This will enable network operators such as BT to offer customers a plethora of exciting new services sourced from a wide range of service providers.

3. Stream-binding solutions

In the same way that the process of allowing two objects in a distributed processing system to exchange information is called ‘binding’, the process of connecting a stream of information between two end-points is called ‘stream binding’.

The OMG describes a stream as representing continuous media transfer between two or more virtual multimedia devices. A stream end-point terminates a stream. A simple example of a stream is a microphone device (audio source or producer) connected to a speaker device (audio sink or consumer) (see Fig 1).

An important feature of a stream connection is that it can use a physically separate transport method to that used for communication between objects. This allows high bandwidth connections to be utilised for the stream information. Figure 2 shows an abstract representation of the ESP, which is based on the principles of the TINA distributed processing environment (DPE) for service management, with a separate connectivity route for streams. The ESP uses CORBA-compliant products such as IONA’s Orbix [8] and Inprise’s Visibroker for Java [9].

The ESP supports a service management platform which is used to offer multimedia services to users in various locations across Europe. The distributed processing environment extends across all the platforms and terminals, and, in keeping with TINA principles, the ESP associates components either with the user or the provider domains. All (computational) objects are interconnected to each other through their respective ORBs via the KTN.

The transport technique for both the KTN and the AV streams is ISDN2, either as separate ISDN connections or by using the same physical connection, as will be discussed later. In order to be able to offer streams-based services, the platform requires a method for stream binding and control, and two models were considered [10], the TINA model and the OMG model.

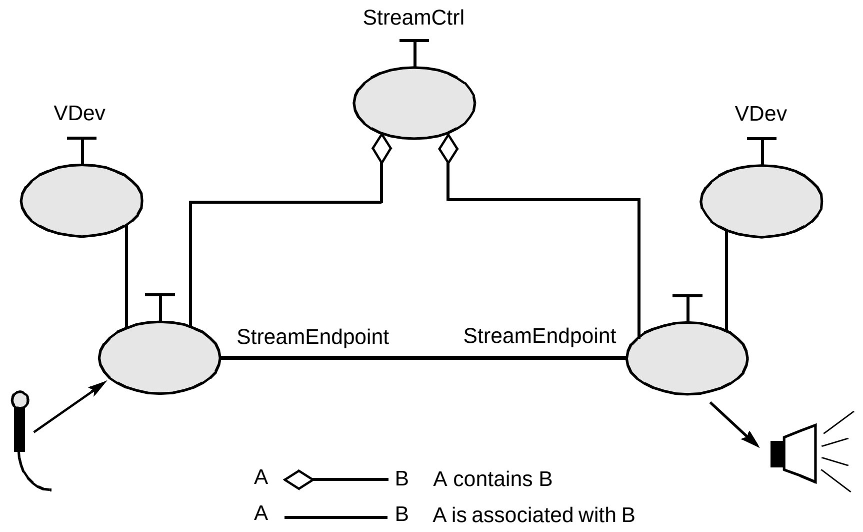

Fig 1 A basic stream configuration.

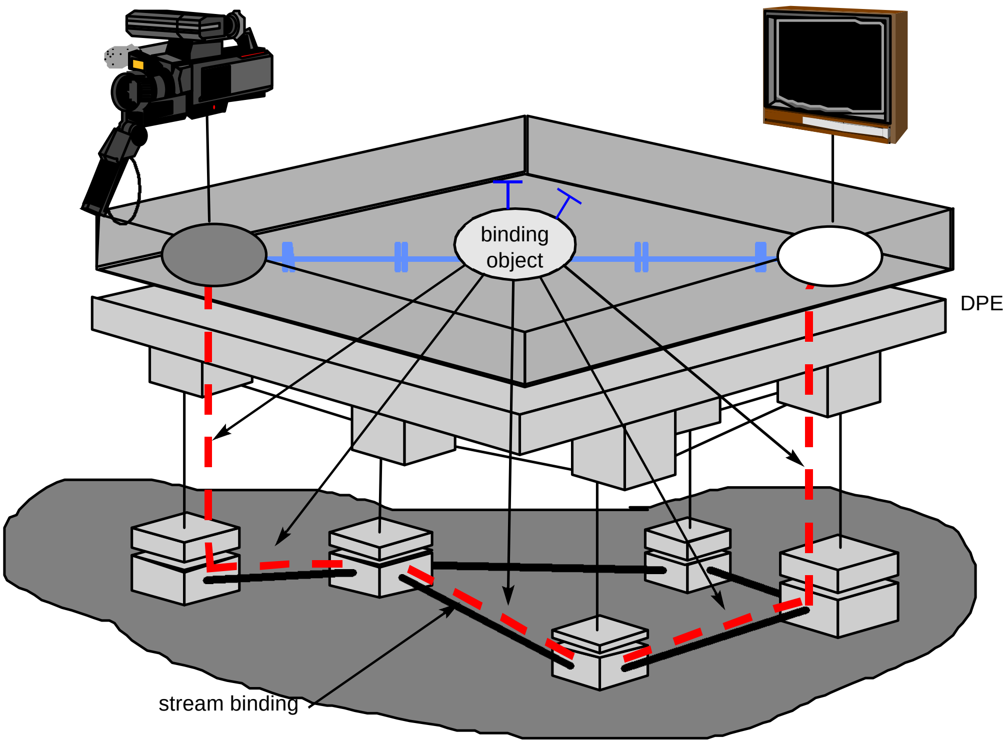

Fig 2 The EURESCOM services platform.

3.1 TINA stream binding

The TINA network resource architecture (NRA) [11] defines a general session model which specifies the associations between the computational objects involved in a particular service. The service session model describes the control and connectivity relations between the members of a service, e.g. end users and service providers. This is called the service session graph (SSG).

3.1.1 Service session graph

Figure 3 shows a simplified service session graph, which represents a general model of possible connectivity relations between stream interfaces and stream bindings that can be established in a service session.

TINA defines the stream-binding session relation class to model complex topologies for time-based flows. It was not possible to express these relations in the logical connection graph (LCG), which was introduced first by the NRA.

Since TINA sub-architectures should complement each other, a relation exists between the SSG and LCG; the stream binding relation in the SSG is supported by one or more LCGs (see Fig 3).

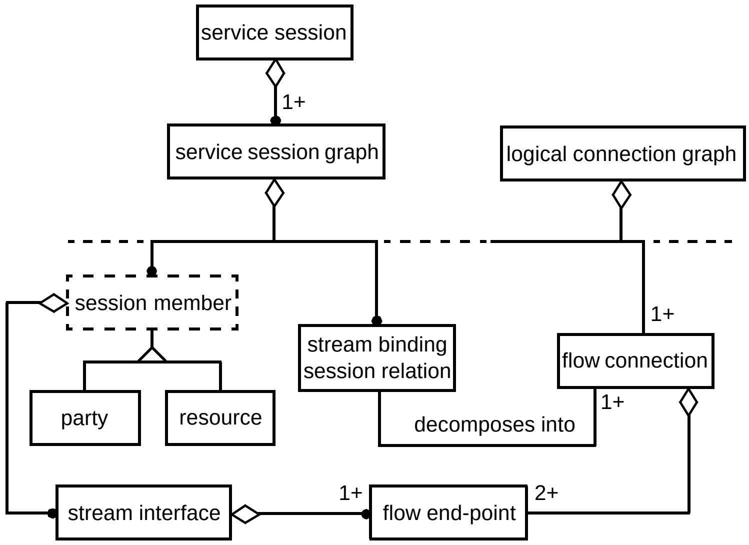

Fig 3 Relation between SSG and LCG to express connectivity.

3.1.2 Logical connection graph

The LCG is introduced in the NRA to express the connectivity between end-points independent of how this is achieved, and independent of the underlying network technology. The LCG is able to express point-to-point bi-directional and point-to-multipoint uni-directional connectivity relations, which is more limited than the SSG, which supports the full range of connectivity topologies described in the NRA.

An LCG is an information object that represents a single stream binding. The LCG contains one or more flow connections. A flow connection is the representation of a binding between a source flow end-point and a number of sink flow end-points.

3.1.3 Nodal connection graph

A nodal connection graph (NCG) is an information object that represents a binding inside the end-user node and it contains one or more nodal connections. A nodal connection is the representation of the connection between a flow end-point and network access point (Figs 4 and 5).

3.1.4 Physical connection graph

A physical connection graph (PCG) is an information object that represents a transport network binding and it

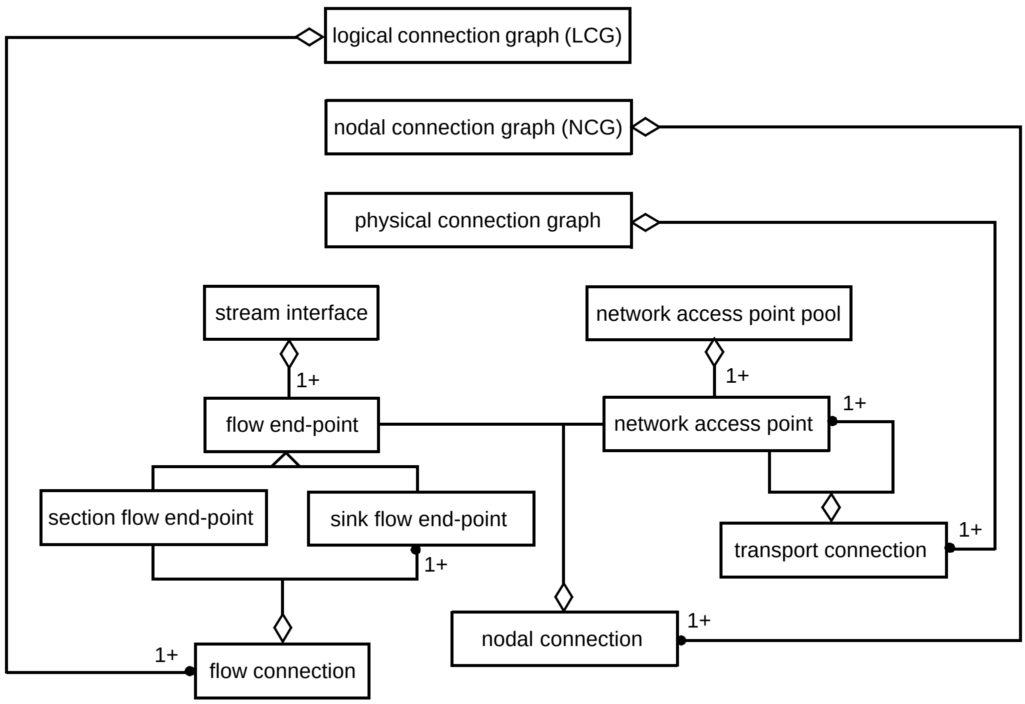

Fig 4 OMT model to express connectivity in TINA.

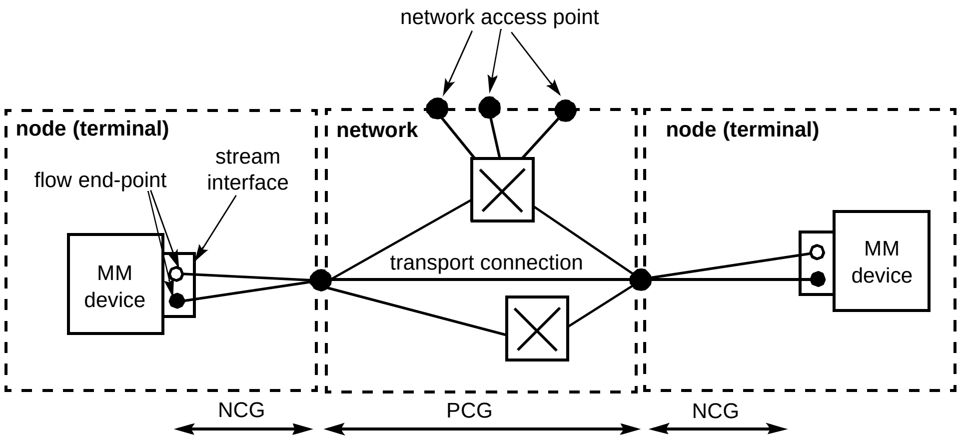

Fig 5 Example of relation between TINA connection graphs.

contains one or more transport connections. A transport connection is the representation of a bi-directional connection between two or more network access points (Figs 4 and 5).

These different connection graphs are used to express connectivity relations for stream interfaces at different granularity levels. The connection graphs are used by different objects, as illustrated in Fig 6.

3.1.5 Concepts related to the computational/engineering viewpoint

TINA defines several generic objects that should be present in a TINA service platform. One of these is the service session manager (SSM) object. The SSM object is the central point of a service session and it co-ordinates and serves all requests from members, who participate in the service session. The SSM object stores the SSG and has the knowledge about connectivity relations between members. It is able to manipulate the SSG through interfaces provided by the SSM object (e.g. removing a member from an ongoing session). The SSM object is also capable of decomposing the SSG into one or more LCGs that are sent to the communication session manager (CSM) object. Manipulation of connectivity relations is achieved by sending updated LCGs to the CSM object, which contains the LCG of a particular session. Additional computational objects are shown in Fig 6 and explained below.

The CSM object is responsible for the end-to-end application communication. It provides the binding between

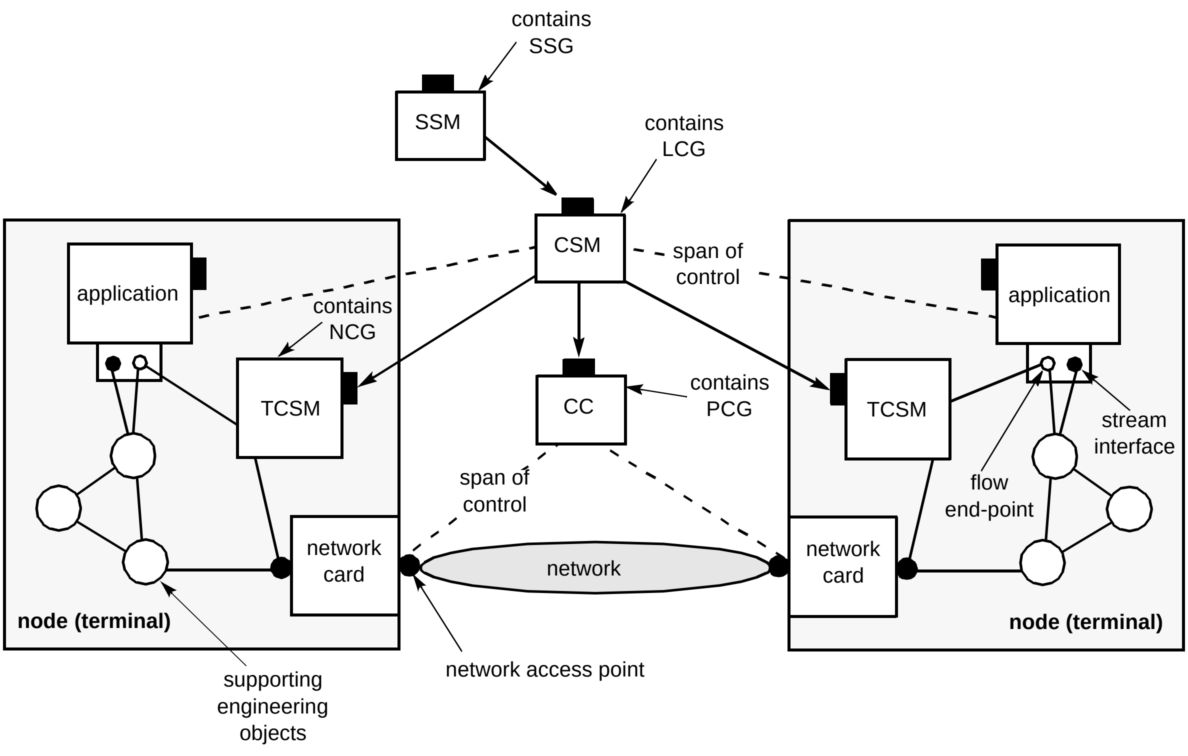

Fig 6 Computational/engineering view for modelling connectivity.

flow end-points (depicted by the dotted lines in Fig 6). The LCG is used by the SSM object to describe and express to the CSM object the characteristics and configuration of the binding between the flow end-points in the context of a communication session. The CSM will then decompose the requested binding (i.e. LCG) into two graphs — the NCG and PCG.

The CSM object requests the terminal communication session manager (TCSM) to take care of the NCG. The TCSM object is responsible for the binding inside a user's node. It binds the flow end-point (typically a multimedia device) to the network access point (typically a network card). The characteristics and configuration of the nodal binding is described by the CSM using the NCG within the context of a terminal communication session.

The characteristics and configuration of the transport connection are described by the CSM by means of the PCG within the context of a transport connection session. The connection co-ordinator (CC) is responsible for the end-to-end transport connection provided by a transport network, i.e. from network access point to network access point.

3.2 OMG stream binding

In October 1997, following the Request for Proposals (RFP) from the OMG on the topic of 'control and management of audio/video streams', IONA Technologies, Lucent Technologies and Siemens-Nixdorf submitted a solution. This specification has now been adopted by the OMG as the 'CORBAtelecoms: Telecommunications Domain Specifications' [12], and consists of a set of CORBA interfaces that implement a distributed media streaming framework for point-to-point and multipoint connections.

Figure 7 shows an example stream architecture from the OMG specification. A StreamEndPoint consists of the following logical components:

- a stream interface control object (1) with an interface specified in OMG's Interface Definition Language (IDL) that allows control and management of a data stream,
- a basic object adapter (2) for message exchange between the ORB and the stream interface control object,
- a flow data end-point (3) which may be a sink or a source,
- a stream adapter (4) for data flow exchange between the network and the flow data end-point,
- StreamEndPoints will be bound and connected with control and management objects (5).

The principal components in the OMG stream control specification are as follows:

- a stream control (StreamCtrl) is responsible for supervising a particular stream binding,
- a multimedia device (MMDev) is a representation of the hardware (microphone, loudspeakers, etc) available at a designated terminal — it is also a factory for stream end-points and virtual devices,
- a virtual device (VDev) is an abstract representation of the hardware — prior to establishing an AVstream, virtual devices at both ends exchange configuration information,
- a stream end-point (StreamEndPoint) represents one end of a stream — both end-points exchange a flow specification,
- flow connections, flow end-points and flow devices are used to manage multiple unidirectional flows within one single stream.

The OMG stream control specification distinguishes between the full profile, which uses all the components mentioned above, and the light profile, in which the flow interfaces are not exposed. A simplified illustration of how a stream is established is shown in Fig 8.

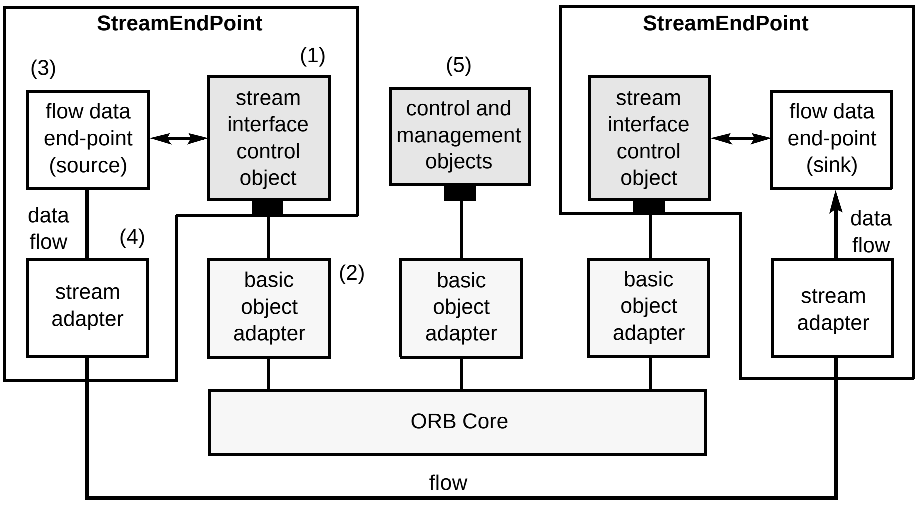

Fig 7 Example stream architecture from the RFP.

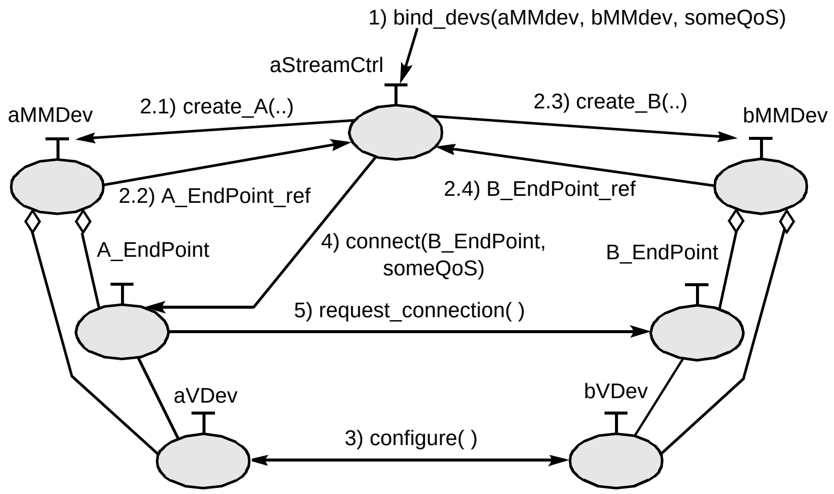

Fig 8 Establishing a stream (simplified).

The sequence of operations required to establish a stream connection between the two MMDevs is illustrated in Fig 8, and is described in more detail in section 4. The StreamCtrl object is in some ways similar to the TINA CSM object, as it is responsible for creating the binding between the two end-points. This object may be situated centrally, e.g. in a retailer domain, while the other objects are situated in the user terminals. The stream connections are set up from end to end (e.g. by dialling an ISDN number) rather than using centralised control of the network.

3.3 Summary

It can be seen that the TINA stream-binding model is based on a complex hierarchy which includes network management aspects such as connectivity over multiple layer networks. The OMG stream control solution, on the other hand, is much more focused on the service layer viewpoint, and provides a better match for the requirements of the ESP platform. There is no emphasis on network control aspects as in the TINA model, and it has the flexibility to be applied with varying levels of control over stream establishment and manipulation. This allows a gradual increase in complexity in the streams implementation work (e.g. starting with a simple point-to-point demonstrator and moving to point-to-multipoint with manipulation of QoS parameters such as bandwidth). The OMG solution has therefore been chosen as the preferred model for stream binding in the ESP.

This work therefore focuses on the integration of the light profile of the OMG stream control model into the TINA environment of the ESP prototype. An explanation of how this was implemented for point-to-point connections is given in section 4.2, and, in section 4.4, multipoint connections are described.

A federated naming service, compliant with the OMG CosNaming specification, is used to store object references of the OMG components. The Visibroker naming service and OmniOrb's OmniNames have been successfully used at different sites and wrapped to appear as one global naming service.

4. Integration of stream binding in the ESP

In the first phase of the project, the main area of work involved assessment of the validity of concepts and ideas developed by the TINA Consortium and the OMG, such as the TINA business model, the TINA reference points and CORBA interoperability. During this first stage, only simple services were required for the demonstration of interoperability between platforms. As the project matured, and with the confidence that TINA and CORBA were proving potentially useful and reliable technologies to BT and partner companies, it was decided to move forward to include advanced services incorporating multimedia features, such as AV streams.

4.1 Design issues

The ESP architecture, in which both TINA and OMG components co-exist, is depicted in Fig 9. Before the stream binding can be established between two clients, an access session as well as a service session must be created between the clients and a retailer.

As described in section 1, true interoperability between heterogeneous systems has been achieved, and it is possible to use the provider agent (PA) developed by one partner to log on to a retailer owned by any other partner.

In order to log on, users need to provide authentication details such as username and password. The PA then contacts the retailer's initial agent (IA) in order to create an access session between the user's PA and the user agent (UA), which is the user representation in the provider domain.

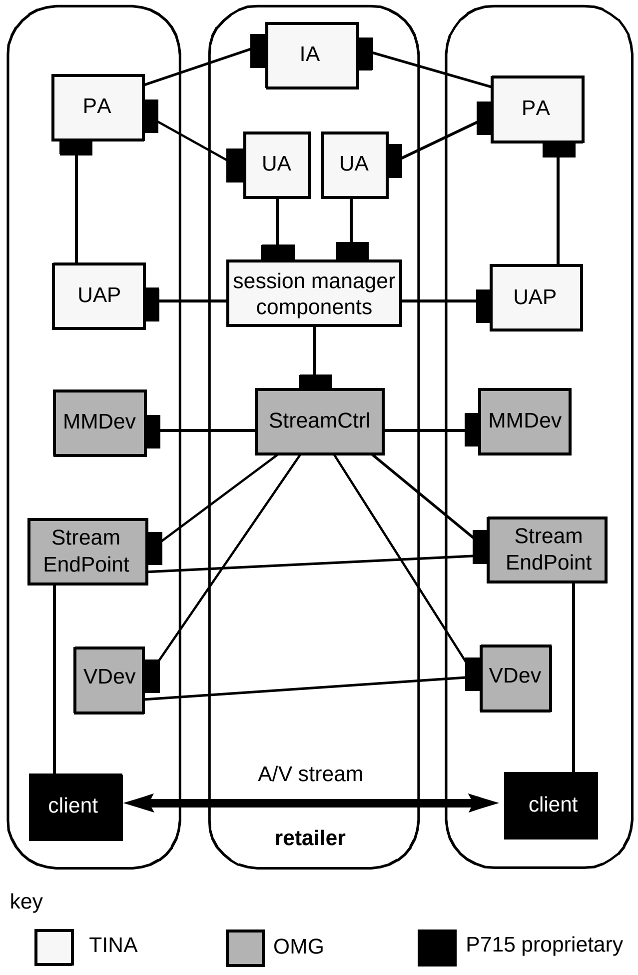

Fig 9 Using audio/video stream control as a communication session in the ESP.

Once this has been created, the first user can retrieve a list of available services from the retailer and ask for the 'videoconferencing' service. A service session is then created. The client-side of the service is downloaded over the network from the retailer to the user's terminal, whereas server-side components (session manager components) are created dynamically by a service factory in the retailer domain.

4.2 Implementation — point to point

The ESP videoconferencing service relies on two ITU-T multimedia standards to implement the stream transport mechanism:

- H.320 [13] for videoconferencing over ISDN,
- H.323 [14] for videoconferencing over TCP/IP.

All the OMG light profile components described earlier have been implemented. Since the main interest of the project is to demonstrate interoperability between heterogeneous systems, the decision was taken to use a range of ORBs, programming languages and conferencing software, as shown in Table 1.

Table 1 Range of systems used in the implementation.

	GUI	Stream Control	Multimedia Device	Conferencing Software
BT	C++ CoolORB	C++ CoolORB	C++ CoolORB	Microsoft NetMeeting
DT/ KPN	Java OrbixWeb	Java Visibroker for Java	C++ CoolORB	Intel Proshare

The first phase of streams implementation involved building a stand-alone OMG stream control demonstrator. At this stage, as there was no link to the ESP, a third party control application presenting a graphical user interface (GUI) was developed. Its role is to retrieve a list of registered clients from the naming service, and to allow a user to choose two clients to participate in a conference. The user may also select the protocol to be used (either H.320 for ISDN-based conferencing or H.323 for TCP/IP-based conferencing) and the bandwidth to be allocated to the AV streams in the case where the H.323 protocol is used. If insufficient resources are available on the network, it is possible to set up an audio-only conference.

Since the OMG stream control specification does not state how the stream control objects are created, a proprietary stream control factory was introduced to create these objects when required.

Before the AV stream can be established, a number of pre-conditions are required:

- the stream control factory and the multimedia device must be launched and running,
- their interoperable object references (IORs) must be available through the naming service, so that they can be contacted by the third party application.

The following is the detailed sequence of operations (see Fig 10) that takes place when an AV stream is established between two designated clients:

- the GUI contacts the stream control factory and asks it to create a new stream control object, whose IOR is returned (1a and 1b),
- the application invokes the `bind_devs` method on the stream control, passing the references to both multimedia devices as parameters (2),
- the stream control asks both multimedia devices to create stream end-points and virtual devices (3a and 3b),

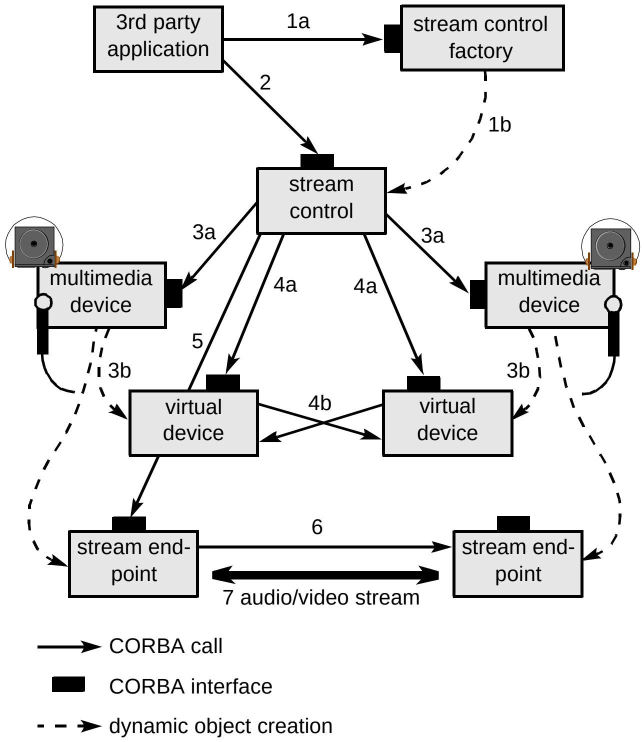

Fig 10 Stream binding operation sequence.

- the stream control will then invoke the set_peer method on each of the two virtual devices, passing the reference to the other one as a parameter, so that each virtual device knows where its counterpart is — each virtual device then invokes the set_format method on its peer, this two-way handshaking protocol (see Fig 11) being used to negotiate the QoS between the two parties (4a and 4b),
- once the QoS has been agreed between the two parties, the stream binding can actually take place — the stream control invokes the connect method on the calling party's stream end-point, passing the reference to the called party's stream end-point as a parameter (5),
- the first stream end-point, upon receiving the reference to the second one, will invoke the request_connect method on the latter, which should then return its ISDN phone number (if the H.320 protocol is used) or its IP address (if the H.323 protocol is used) (6),
- once the calling party's stream end-point receives the called party's contact details, the connection can be established — the calling party's stream end-point will contact the underlying conferencing software and ask for the called party's stream end-point to be called (7),
- when the binding is due to terminate, the unbind method is invoked on the stream control, which will then:

  — instruct the stream end-points to cease the conference,

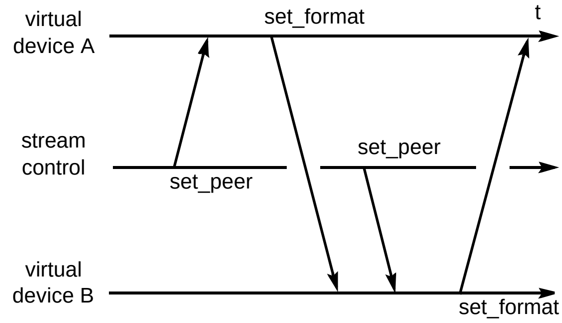

Fig 11 Two-way handshaking.

  — tell the multimedia devices to destroy unnecessary components (virtual devices and stream end-points).

Integration of this stream control functionality with the ESP was fairly straightforward, but required some specially designed TINA components to be developed.

On the client side, the reference to the multimedia device, running on the client terminal, has to be retrieved from the naming service and passed to the retailer in a standardised manner, which must be agreed between partners.

On the retailer side, the reference is forwarded to the server-side components, which must be able to handle references to several clients in the same session. These components are then responsible for contacting the stream control and establishing the relevant stream bindings.

The following section outlines this integration in more detail and describes a typical scenario.

4.3 *Example scenario*

An illustration of how the implementation works in practice is given by a typical scenario for two users accessing a retailer to use the videoconferencing service.

- When the user starts a new conference, the provider agent retrieves the reference to the user's multimedia device from the naming service. This is effected by the provider agent knowing the user's name (as part of the authentication details), which is also the identifier used by the multimedia device to register with the naming service. The PA then sends the retrieved IOR to the retailer through the Ret reference point.
- The retailer then sends the multimedia device's IOR, which it has just obtained from the PA, to the session manager components (see Fig 12).
- The user, by means of the UAP GUI, can then invite a second user to join the conference. Currently, the user to be invited must have already logged on to the same

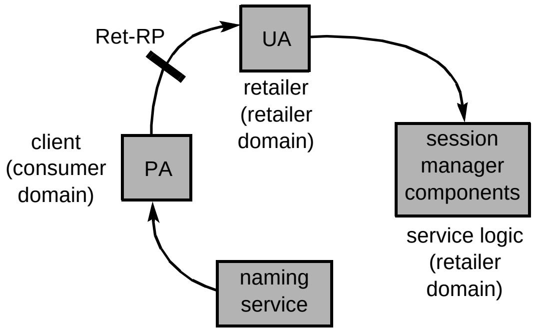

Fig 12 Information flows in the service set-up phase.

retailer. One direction of future work is implementation of the third party reference point (3Pty-RP) and the retailer-to-retailer reference point (RtR-RP), which will allow users to interact with each other even when they are logged on to different retailers.

- The invitation will be issued by the first user's UAP and then sent to the retailer. At this stage, the retailer will contact the invited user, through their PA, advising where the invitation is from and for which service. It is now down to the second user either to accept the invitation and join the service session, or to refuse it.

- Assuming the second user accepted the invitation, the client-side of the service (UAP) will be downloaded over the network from the retailer and, using the UAP GUI, the second user can join the existing conference.

- Using the same method as for the first user, the second user's multimedia device IOR is retrieved from the naming service by their PA and sent to the retailer and thence to the server-side code.

- Once the session manager components receive the references to two multimedia devices, the stream binding can actually take place. Two scenarios are possible, depending on whether the point-to-point or the multipoint approach is used:

  — if the service has been configured to work in the point-to-point mode, the session manager components will establish a stream binding between the two multimedia devices, by making a call to the stream control object as described in section 3,

  — if the server has been configured to work in the multipoint mode, the session manager components will establish a stream binding between each multimedia device and a multipoint control unit (MCU) — in this particular scenario, it is then possible to invite more users to join the conference (a full explanation of multiparty stream binding is given in the next section).

Any user can, at any time, decide to end their participation through their own UAP GUI. The first user, as the initiator of the conference, can decide to terminate the conference, resulting in everybody, including that user, being disconnected and forced to leave the service session.

4.4 Multiparty stream binding

The point-to-point stream binding as explained above only allows connectivity between two users. In order to bind more than two users together, multiparty stream binding is required. Multicast technology could be used to distribute the stream data between all parties, but would require specially equipped routers to be available at all points in the network. Multicast offers advantages in network bandwidth utilisation, and will be the subject of further study to evaluate its use for implementation of multiparty stream bindings.

An alternative method is to use a central mixing unit, known as a multipoint control unit (MCU), and this is the approach adopted in the ESP. Depending on the application, MCUs come in two types, hardware and software. Hardware MCUs are widely used for multiparty (H.320) videoconferencing, but are very expensive to buy. With the advent of IP-based conferencing using the H.323 standard, software MCUs have started to become available at a fraction of the cost of their hardware counterparts, and the ESP has chosen an H.323 MCU for implementation of its multiparty conferencing application.

Arbitrary topologies can be formed using MCUs and H.323 terminals. In the current implementation, only a single-level topology is supported, with a unique MCU and attached H.323 terminals. Future extensions will allow support for more complex topologies with multiple MCUs, such as the one shown in Fig 13.

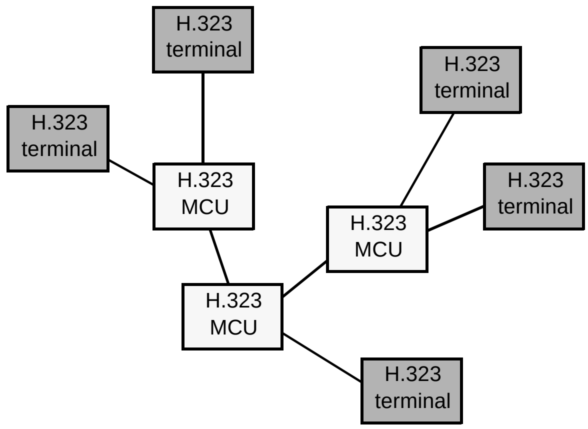

Fig 13 Sample topology of H.323 MCUs and H.323 terminals.

A multiparty stream binding for *N* users can be considered as a set of *N* point-to-point stream bindings between each user terminal and a unique MCU. The interfaces for the multiparty stream binding are therefore almost the same as for the point-to-point case. Only the interpretation (operational semantic) of the `bind_devs` request had to be extended to allow the specification of the MCU (first invocation) and all client terminals (subsequent calls).

A new operation, unbind_party, has been added to allow the removal of a single terminal from the multiparty stream binding.

Figure 14 shows the computational objects involved in a multiparty stream binding.

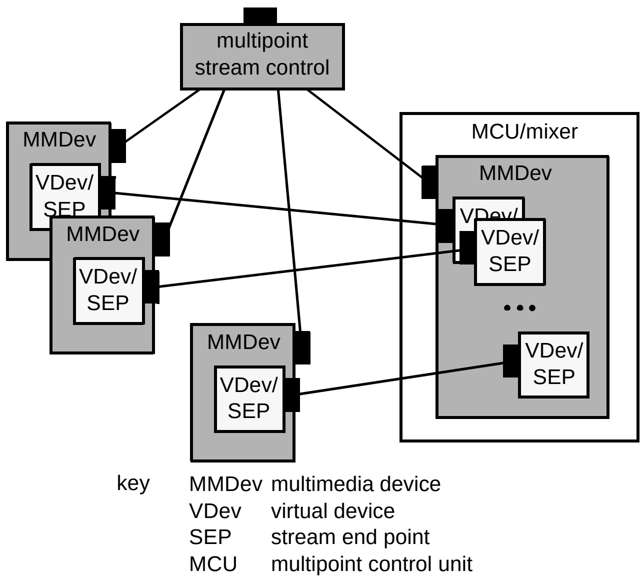

Fig 14 Using an MCU mixer for creating multipoint-to-multipoint streams.

5. Results and evaluation

Interoperability tests have been run between the six European sites to verify that users at each site can request and run services offered by retailers at all other sites. At the time of writing, five out of six sites have proved interoperability of the TINA platforms across the Ret interface, with the last site being in the process of upgrading to the latest Ret specification. Currently just two partners, BT and DT, have implemented the OMG streams integration, and tests have been run successfully to show that a videoconference service can be requested and set up between the two sites.

During the implementation of both the EURESCOM services platform and the videoconferencing service, some difficulties were encountered, most of which can be attributed to interoperability problems between the various ORB products that were used.

A conclusion that can be drawn is that these ORB products are not strictly CORBA 2.0 compliant. This was overcome, either by slightly changing the design approach, or more often by upgrading the ORB products following discussion with the vendors.

Some further findings from the tests are as follows.

- Quality of service (QoS)

  When using the H.320 protocol (ISDN-based conferencing) in a point-to-point conference, both the audio and video quality are found to be very good. This is due to the 64 kbit/s throughput available on a typical ISDN B channel. Using the H.323 protocol (TCP/IP-based conferencing) in a point-to-point connection, a minimum bandwidth of 48 kbit/s is necessary to obtain an acceptable quality of audio and video.

  When using the meeting point MCU in a multipoint H.323 conference, the audio stream requires a steady 10 to 15 kbit/s bandwidth, whereas the video stream varies between 3 and 30 kbit/s. The reason for this is that the video codec implemented by the MCU only sends the parts of the picture that have changed, rather than the whole picture. The resulting utilisation of the network is therefore quite dependent on whether the picture is still or moving.

- Scalability

  The ESP has been developed as a prototype platform, and no extensive study has been conducted so far on the topic of scalability. Future projects will investigate this issue. However, to set up a new stream binding only requires a few messages between the various TINA and OMG components, each of which accounts for no more than a few hundred bytes, so the overall overhead is not likely to present a problem for scalability.

  There is a possibility that the MCU could become overloaded if too many clients try to participate in a conference at the same time. One solution to this which will be investigated is the federation of MCUs, which will allow the building of more complex topologies, but will also require development of the necessary management tools.

6. Conclusions and future work

EURESCOM project P715 has successfully achieved the integration of TINA service architecture components with OMG stream-binding functionality, using a standards-based pan-European demonstrator to offer services to users in six different countries. Although currently only two companies have implemented the full range of services to include stream-based offerings, such as videoconferencing, the feasibility of this technology has been proven, and it would be a relatively simple task to port the stream-binding functionality to the other platforms.

This work has demonstrated the feasibility of using a TINA-compliant service management platform in conjunction with a standards-based stream-binding technique and commercially available products to offer multimedia services to users on a global scale. The use of the TINA specifications to provide a standard open interface for true interoperability at the service and application level has also been proved.

Future work in forthcoming projects will involve the integration of multipoint videoconferencing and the investigation of management issues such as fault and performance management. Future work on streams will undoubtedly focus on ways to guarantee an agreed QoS. This could be achieved by using such protocols as the reservation protocol (RSVP) or using a network layer such as ATM, which provides the ability to reserve a predefined bandwidth.

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References

1 TINA-C Web site — http://www.tinac.com

2 ISO/IEC 10746-3: 'Open Distributed Processing — Reference Model (RM ODP), Part 3, Architecture', (1995).

3 Object Management Group: 'The Common Object Request Broker: Architecture and Specification, Version 2.2', (February 1998).

4 EURESCOM Web site — http://www.eurescom.de

5 Ainslie B J, Fisher M A and Rana S: 'TINA — a distributed approach to service management', BT Technol J, 17, No 2, pp 95—102 (April 1999).

6 Mungee S, Surendran N and Schmidt D C: 'The design and performance of a CORBA audio/video streaming service', HICSS-32 International Conference on System Sciences, Hawaii (January 1999).

7 Rana S P, Fisher M A and Egelhaaf C: 'Implementation and interoperability experiences with the TINA Service Management Specification', TINA Conf (TINA'99), Hawaii (April 1999).

8 IONA Web site — http://www.iona.com/products/index.html

9 Inprise Web site — http://www.inprise.com/visibroker

10 EURESCOM: 'P715 Deliverable D2, Annex 4-1: State of the Art Regarding Streams and Binding Objects, Version 1.0', (March 1999).

11 TINA-C: 'Network resource architecture, TINA Baseline, Version 3.0', (February 1997).

12 Object Management Group: 'CORBAtelecoms: Telecommunications Domain Specifications', Version 1.0 (June 1998).

13 ITU-T Recommendation H.320: 'Audiovisual and multimedia systems, Systems and terminal equipment for audiovisual services, Narrow-band visual telephone system and terminal equipment', (July 1997).

14 ITU-T Recommendation H.323: 'Audiovisual and multimedia systems, Systems and terminal equipment for audiovisual services, Packet based multimedia communications systems', (February 1998).

Peter Loosemore joined BT in 1972 as an apprentice in Birmingham, and subsequently moved to BT Laboratories in 1975, where he worked on component reliability and semiconductor device fabrication, and was a member of a team which produced a number of world-leading devices. In 1994 he led a team working on a European collaboration project to investigate the use of screen-phones for IN services. Since then he has taken a leading role in a number of collaboration projects on future service and network architectures.

Eric Sellin joined BT Laboratories in September 1997 after obtaining a post-graduate engineering degree in Information Technology from the University of Rennes, France.

He is currently with the Distributed Computing and Information Systems group, working on the development of end-user services for a pan-European services management platform, with a particular interest in the control and management of audio/video streams. He is an associate member of the IEE.

Sohail Rana gained a first class honours degree in Electrical and Electronics Engineering in 1994 and was presented with the IEE award. He worked for Lockheed Martin UK for a year and went on to study for an MSc in Data Communication Systems.

He joined BT in 1996 as a graduate, and is currently working on a pan-European project, (EURESCOM P715), which has involved implementation of a TINA services manage-ment platform. He is also studying for the BT MSc.